Assessment 8: K-Means Clustering

1. Load the dataset shared through e-mail.
2. View the dataset
3. Check missing values
4. Check for the dependent variable distribution. (table)
5. Select only numerical features for K-Means clustering (Others remove it).
6. Standardize the data. (scale)
7. Apply K-Means Clustering (kmeans) for center as 2.
8. Display K-Means results
9. Display cluster centers
10. Calculate WCSS for K = 1 to 10
11. Display the elbow plot.
12. Apply K-means clustering based on the elbow recommendation.

Hierarchical Clustering

1. Load the dataset shared through e-mail.
2. View the dataset
3. Check missing values
4. Select only numerical attributes
5. Standardize the data (scale)
6. Calculate Euclidean distance (dist)
7. Perform Hierarchical Clustering using Ward's method (hclust)
8. Plot the dendrogram. (plot)
9. Divide the data into 3 clusters and Display cluster assignments(cutree)
10. Display number of samples in each cluster. (table)